From Health Indicators to Health Awareness: A Comparative Study of Traditional Machine Learning and Deep Learning for Three-Class Diabetes Classification

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GitHub repository	https://github.com/Janviere-dev/diabetes-risk-prediction
Google Drive folder	https://drive.google.com/drive/folders/1B_ASskZvPqFe79gLaTGVG7Myz0Eewpig?usp=sharing
Weights and Biases project	https://forge.coreweave.com/wandb/janvieremunezero16-alu-african-leadership-university-/diabetes-health-indicators?nw=nwuserjanvieremunezero16
Presentation video	Demo video URL link is in the README file on GitHub.

Abstract

This project compares traditional machine-learning methods and deep neural networks for classifying diabetes-related status from public health indicators. The motivation is personal: a family member lives with diabetes, and this led me to explore whether a future mobile tool could help people notice when professional screening may be worth seeking. The current study does not build a clinical diagnostic system. It uses the Diabetes Health Indicators dataset derived from the 2015 U.S. Behavioral Risk Factor Surveillance System, with 253,680 records, 21 predictors, and three target classes: no diabetes, prediabetes, and diabetes. The original target distribution was strongly imbalanced, with 84.24% of records in the no-diabetes class and only 1.83% in prediabetes. Eight traditional-model experiments and seven neural-network training experiments were conducted, followed by a validation-only decision-threshold analysis. Group-based splitting reduced leakage from identical predictor patterns across partitions. The strongest traditional model by validation macro F1 was a balanced Random Forest (0.4428), while the deeper MLP with focal loss reached 0.4551. On the untouched test set, the MLP obtained 0.4536 macro F1 and 82.15% accuracy. Applying a validation-selected prediabetes threshold increased test macro F1 to 0.4614, but prediabetes precision remained 3.90% and recall 6.28%. A separate cleaning sensitivity study removed duplicate records and conflicting label patterns; performance changed only slightly. These results show that overall accuracy can hide a serious minority-class weakness. More representative data, independent validation, and further work on prediabetes detection are needed before the approach could support a health-awareness application.

Index Terms: Diabetes classification, class imbalance, traditional machine learning, deep learning, focal loss, health indicators, precision-recall analysis, sensitivity analysis.

I. INTRODUCTION

Diabetes is a major public-health concern. The World Health Organization estimated that about 830 million people were living with diabetes in 2022, with prevalence increasing faster in low- and middle-income countries than in high-income countries [1]. Prediabetes is also important: blood glucose is higher than normal but not high enough for a type 2 diabetes diagnosis, and many people do not know they have it. The U.S. Centers for Disease Control and Prevention reports that eight in ten adults with prediabetes are unaware of their condition [2]. These figures illustrate why health awareness and screening matter.

My interest is personal because someone in my family lives with diabetes. This made me think about how a person could become more aware of their health and be encouraged to seek screening. I imagined a simple mobile application where users could enter basic health information and receive a preliminary indication that a checkup may be worthwhile. It would not tell someone they definitely have diabetes or replace doctors and laboratory testing; it would only be investigated as a prompt to seek professional advice.

That long-term idea is broader than this experiment. The dataset contains survey-reported diabetes-related categories, not laboratory-confirmed labels or measurements designed to predict future disease onset. The technical question here is therefore narrower: how well can traditional machine-learning models and neural networks classify no diabetes, prediabetes, and diabetes from available health indicators?

The project uses Alex Teboul's Diabetes Health Indicators dataset on Kaggle, derived from the 2015 Behavioral Risk Factor Surveillance System (BRFSS) [3], [4]. The selected file contains 253,680 rows, 21 predictors, and one target. Features include age, BMI, blood pressure, cholesterol, physical activity, general health, and other demographic or behavioural indicators.

The original target distribution was highly imbalanced: 84.24% no diabetes, 1.83% prediabetes, and 13.93% diabetes. Accuracy can therefore look strong while a model misses prediabetes. This project compares macro F1, class-specific precision and recall, ROC-AUC, and average precision alongside accuracy, and uses learning curves and confusion matrices to investigate error patterns.

This project compares eight traditional machine-learning experiments with seven neural-network training experiments, followed by validation-only threshold tuning. It evaluates selected candidates on held-out test data and separately examines cleaning records with duplicates or conflicting labels. The report is organized into literature review, methodology, results, discussion, and conclusion.

II. LITERATURE REVIEW

A. Diabetes risk assessment and the purpose of screening

Earlier diabetes risk research shows both the value of health indicators and the need for population-specific validation. Hippisley-Cox et al. developed and validated QDScore to estimate ten-year risk of diagnosed type 2 diabetes using routinely collected information from primary-care cohorts [5]. That is different from the present project, which classifies survey reported status at one point in time rather than estimating future disease onset.

Noble et al. reviewed type 2 diabetes risk scores and found variation in development, validation, and readiness for practice [6]. Collins et al. also identified weaknesses in reporting outcome definitions, predictors, missing data, and model performance [7]. Together, these reviews support transparent reporting and caution against choosing models by headline accuracy alone.

Chang et al. evaluated machine-learning methods using health indicators and reported promising Random Forest results [8]. Their study is relevant because it uses a similar survey-indicator setting. However, headline scores from separate studies are not directly comparable because splits, class proportions, preprocessing, and metrics differ. This project instead compares methods under one documented split and gives macro F1 and minority-class recall particular attention.

B. Machine learning in diabetes prediction and validation

De Silva et al. reviewed machine-learning prediction models in community settings and reported promising discrimination, but also methodological and reporting concerns [9]. None of the included studies reported external validation, and only a minority assessed calibration. The lesson for this project is that a good held-out test score is internal evidence, not proof that a model will work in another population or a real application.

The TRIPOD statement emphasizes transparent reporting so readers can judge prediction-model methods and intended use [10]. Although this project is not a clinical model study, the principle is useful: dataset provenance, target definition, splits, preprocessing, configurations, and failure cases must be clear. The notebooks record experiments and metrics in Weights and Biases (WandB) and CSV files.

The gap addressed here is narrow: under one group-based split and preprocessing pipeline, how do traditional methods and MLP variants trade off majority-class performance against prediabetes detection? A separate sensitivity notebook also tests whether duplicate and conflicting-label removal changes results. This is a reproducible comparison, not a claim of clinical novelty.

C. Class imbalance and evaluation metrics

Class imbalance can lead learners to favour the majority class because doing so reduces overall error. He and Garcia review the methods and evaluation principles needed for imbalanced data [11]. Here, no diabetes accounts for more than four-fifths of records, while prediabetes is below two percent.

Class weighting changes the cost of errors, focal loss emphasizes poorly classified examples, and resampling changes the examples seen during training. Chawla et al. introduced SMOTE to synthesize minority examples [12]. SMOTE was not used in this project; instead, class weighting, focal loss, and random oversampling were tested. Random oversampling performed poorly, showing that balanced class counts do not guarantee useful decision boundaries. Any future SMOTE/SMOTENC experiment would need to preserve valid binary and ordinal feature values.

Evaluation metrics, ROC analysis compares true-positive and false-positive rates across thresholds [13], but Saito and Rehmsmeier argue that precision-recall plots are especially informative for strongly imbalanced data because precision shows the proportion of positive predictions that are correct [14]. In this study, prediabetes ROC-AUC was higher than hard-label F1 might suggest; this is possible because AUC measures ranking across thresholds, whereas a confusion matrix shows decisions at one threshold. Low average precision still signals weak practical detection.

Algorithms have different inductive biases. Random Forest combines many trees [15], gradient boosting builds an additive model in stages [16], and CatBoost uses ordered boosting and specialized categorical-feature handling [17]. An MLP learns nonlinear mappings through layered representations [18]. Greater complexity does not guarantee better results, so algorithm and hyperparameter choices were compared empirically.

D. Research gap and connection to this project

The literature supports health-indicator modelling but cautions against treating internal performance as proof of a deployable screening tool. This project compares traditional and neural methods on an imbalanced three-class dataset, then evaluates threshold tuning and cleaning effects.

III. METHODOLOGY

A. Dataset source and target definition

The study used the file (diabetes_012_health_indicators_BRFSS2015.csv) from the Diabetes Health Indicators dataset published by Alex Teboul [4]. The dataset is based on the CDC's 2015 BRFSS survey, which includes telephone survey responses across U.S. states and selected territories [3]. The raw selected file contained 253,680 observations and 22 columns: 21 input features and the target Diabetes_012. The target was treated as a three-class categorical variable: 0 for no diabetes, 1 for prediabetes, and 2 for diabetes, following the dataset's coding.

The predictors combine binary indicators, ordered categories, and numerical measures. Binary features include high blood pressure, high cholesterol, cholesterol check, smoking, stroke, heart disease or attack, physical activity, fruit and vegetable consumption indicators, heavy alcohol consumption, healthcare coverage, cost-related lack of care, difficulty walking, and sex. Ordered features include general health, age category, education, and income category. BMI, days of poor mental health, and days of poor physical health were treated as numerical inputs for scaling. The retained predictors are survey indicators, not laboratory measurements.

B. Data quality assessment and exploratory analysis

The audit found no missing values encoded in the selected file, although this does not rule out reporting noise or label ambiguity. BMI ranged from 12 to 98 and was right-skewed, with a small number of unusually high values. These records were retained because unusual values may be valid; the EDA included target counts, BMI plots, a correlation heatmap, and class proportions by high blood pressure and high cholesterol. Correlations were treated as associations, not evidence of causation.

The audit found 23,899 exact duplicate rows (9.42%). It also found 1,834 predictor patterns involving 6,120 records with conflicting labels. The primary workflow retained them but kept identical predictor patterns within one partition. A separate sensitivity notebook removed conflicting-label patterns and then exact duplicates so that cleaning effects could be reported separately.

C. Group-based splitting and prevention of leakage

The original workflow used a group-based split with random state of 42. Groups were defined by all predictor columns, excluding the target. Approximately 15% of groups were reserved for testing, then approximately 15% of the full dataset for validation. The resulting partitions contained 177,251 training, 38,203 validation, and 38,226 test rows, with similar but not identical class proportions. No identical predictor-group IDs crossed partitions.

This strategy reduces leakage from identical predictor patterns across partitions, but it does not remove duplicates within a partition or establish full statistical independence. The test set remained untouched for model selection. Threshold 0.24203 was selected on validation probabilities only and then held fixed for test evaluation.

D. Preprocessing and input pipeline

A ColumnTransformer passed through 14 binary indicators and standardized four ordinal and three numerical features. The scaler was fitted on training data only, then applied to validation and test data. No feature-selection algorithm was applied; all 21 predictors were retained because they were available and relevant. A future ablation study could test whether a smaller feature set performs as well.

For deep learning, processed arrays were converted to float32 and loaded through TensorFlow's tf.data pipeline: shuffled training examples (buffer 50,000), batch size 512, and prefetching. Validation data was batched and prefetched without shuffling. Random seed 42 was used. Training ran for up to 30 epochs with early stopping and learning-rate reduction, on CPU-only hardware.

E. Models and experiment design

Eight traditional experiments were conducted. Experiments 1 to 2 compared Logistic Regression with and without balanced class weights. Experiments 3 to 4 compared a balanced 200-tree Random Forest (minimum leaf size 2) with a constrained version (depth 20, leaf size 5). Experiments 5 to 6 compared Gradient Boosting (100 estimators, learning rate 0.1, depth 3) with and without balanced sample weights. Experiments 7 to 8 used CatBoost (500 iterations, learning rate 0.05) at depths 6 and 8, with balanced class weights.

Multi-Layer Perceptrons(MLPs) were used because the 21 inputs are structured health indicators, not images or sequences. Experiment 9 used a [128,64] network with balanced weights; Experiment 10 removed weights; Experiment 11 used inverse-square-root weights; and Experiment 12 introduced focal loss with gamma 2. Experiment 13 increased the layers to [256, 128, 64] with dropout [0.3, 0.3, 0.2]. Experiment 14 used random oversampling, while Experiment 16 added Batch Normalization and L2 regularization (0.0001) to the deeper architecture. MLPs used Adam at 0.001 and early stopping.

Experiment 15 changed the decision rule rather than training a new model. It selected a threshold from Experiment 13's validation probabilities. When the prediabetes probability was at least the threshold, class 1 was predicted; otherwise, the model chose between classes 0 and 2. The selected threshold was 0.24203 and was not adjusted using test data.

F. Evaluation metrics and experiment tracking

Accuracy is the fraction of correct predictions, while macro F1 weights the three classes equally and weighted F1 accounts for class support. Per-class precision and recall describe false-positive and false-negative behaviour. Multiclass ROC-AUC was calculated one-vs-rest, and Average Precision summarized each class's precision-recall performance. Confusion matrices show error counts. Learning curves were interpreted alongside these metrics because ordinary cross-entropy, class-weighted loss, and focal loss have different scales.

Configurations and metrics were tracked in Weights abd Biases, with selected histories and tables saved as CSV files. Model choice was based on validation macro F1 before final testing. The final original-data comparison used the balanced Random Forest, deeper MLP, and the MLP with the fixed validation-selected threshold. Cleaning results were recorded separately.

IV. RESULTS

A. Dataset distribution and preprocessing

The original target distribution was highly imbalanced. Of 253,680 records, 213,703 were labelled no diabetes (84.24%), 4,631 prediabetes (1.83%), and 35,346 diabetes (13.93%). The imbalance was not just a data description; it was visible in the outcomes of the models. Several models achieved accuracy above 84% while never predicting prediabetes. Fig. 1 compares the original target distribution with the distribution remaining after the later sensitivity cleaning. The scaled partitions contained 21 features

each, and no non-finite values were found in the processed arrays.

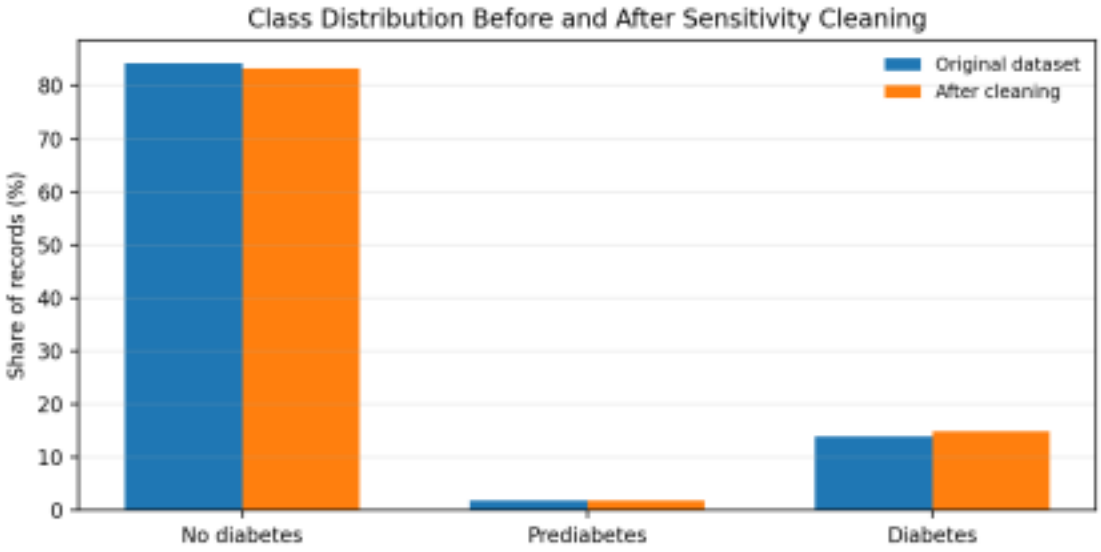


Fig. 1. Class proportions in the original dataset and after the separate duplicate/conflicting-label sensitivity cleaning.

B. Traditional machine-learning experiments

The traditional experiments show a consistent tension between overall accuracy and minority-class recall. The Logistic Regression and Gradient Boosting baselines achieved accuracy of 84.54% and 84.83%, respectively, but their prediabetes recall was zero. Balanced Logistic Regression reduced accuracy to 64.75% while improving prediabetes recall to 29.07% and diabetes recall to 58.90%. In this context, the reduction in accuracy reflects a changed decision balance: fewer majority-class predictions and more predictions for the minority classes, some of which were false positives.

The balanced Random Forest had the highest traditional-model validation macro F1 (0.4428). Its diabetes recall was 67.82%, but prediabetes recall was only 1.39%. Constraining tree depth increased diabetes recall to 73.13% but reduced macro F1 to 0.4327. Weighted Gradient Boosting and balanced CatBoost improved prediabetes recall to 34.08% and 26.84%, respectively, while reducing accuracy and creating more false positives. Increasing CatBoost depth from 6 to 8 slightly raised macro F1 but lowered prediabetes recall.

TABLE I. VALIDATION RESULTS FOR TRADITIONAL MACHINE-LEARNING EXPERIMENTS

Exp.	Model / change	Val. accuracy	Macro F1	Prediabetes recall	Diabetes recall
1	Logisti Rewgression Baseline	0.8454	0.3936	0.0000	0.1779
2	Logistic Regression balanced	0.6475	0.4281	0.2907	0.5890
3	Random Forest balanced	0.7583	0.4428	0.0139	0.6782
4	Random Forest tuned	0.7146	0.4327	0.0403	0.7313

5	Gradient Boosting baseline	0.8483	0.4021	0.0000	0.1960
6	Gradient boosting weighted	0.6232	0.4230	0.3408	0.6161
7	CatBoost balanced	0.6387	0.4231	0.2684	0.6260
8	CatBoost tuned	0.6659	0.4296	0.1864	0.6780

C. Neural-network experiments and threshold tuning

The neural-network experiments showed similar trade-offs. The unweighted MLP reached 84.85% validation accuracy but only 0.3913 macro F1 and no prediabetes predictions. Balanced weights lowered accuracy; inverse-square-root weights raised macro F1 to 0.4517 but still yielded zero prediabetes recall. Focal loss reached 0.4543 macro F1, also without hard prediabetes predictions.

Increasing the hidden layers from [128, 64] to [256, 128, 64] in Experiment 13 produced the best macro F1 among trained neural networks on validation data, 0.4551, with diabetes recall of 53.20%. The additional regularization in Experiment 16 produced the same macro F1 to four decimal places and slightly higher accuracy, but diabetes recall fell to 50.53%, and prediabetes recall remained zero. This suggests that the deeper architecture and focal-loss setup was a slightly better candidate for the project objective, even though neither network was adequate for prediabetes detection.

Experiment 14 was a negative result. Random oversampling expanded the training set to 447,783 records with equal class counts. Validation accuracy fell to 46.62% and macro F1 to 0.3449. Prediabetes recall reached 21.14% and diabetes recall 83.31%, but prediabetes precision was only 1.68%, and no-diabetes recall fell to 41.07%. The unstable, high validation loss indicates poor fit to the original validation distribution; loss values need care because the class distribution changed during training.

TABLE II. VALIDATION RESULTS FOR NEURAL-NETWORK EXPERIMENTS AND THRESHOLD ANALYSIS

Exp.	Model / change	Val. accuracy	Macro F1	Prediabetes with precision-recall	Diabetes recall
9	Baseline MLP with balanced weights	0.6613	0.4260	0.0306 / 0.1446	0.7557
10	MPL without weights	0.8485	0.3913	0.0000 / 0.0000	0.1656
11	MPL with adjusted class weights	0.8212	0.4517	0.0000 / 0.0000	0.5010
12	MLP with focal loss	0.8237	0.4543	0.0000 / 0.0000	0.5051
13	Deep MLP with focal loss	0.8194	0.4551	0.0000 / 0.0000	0.5320
14	MLP with Random oversampling	0.4662	0.3449	0.0168 / 0.2114	0.8331
15	Experiment 13 and tuned threshold	0.8027	0.4684	0.0497 / 0.0793	0.4911

16	Regularized MLP	0.8248	0.4551	0.0000 / 0.0000	0.5053

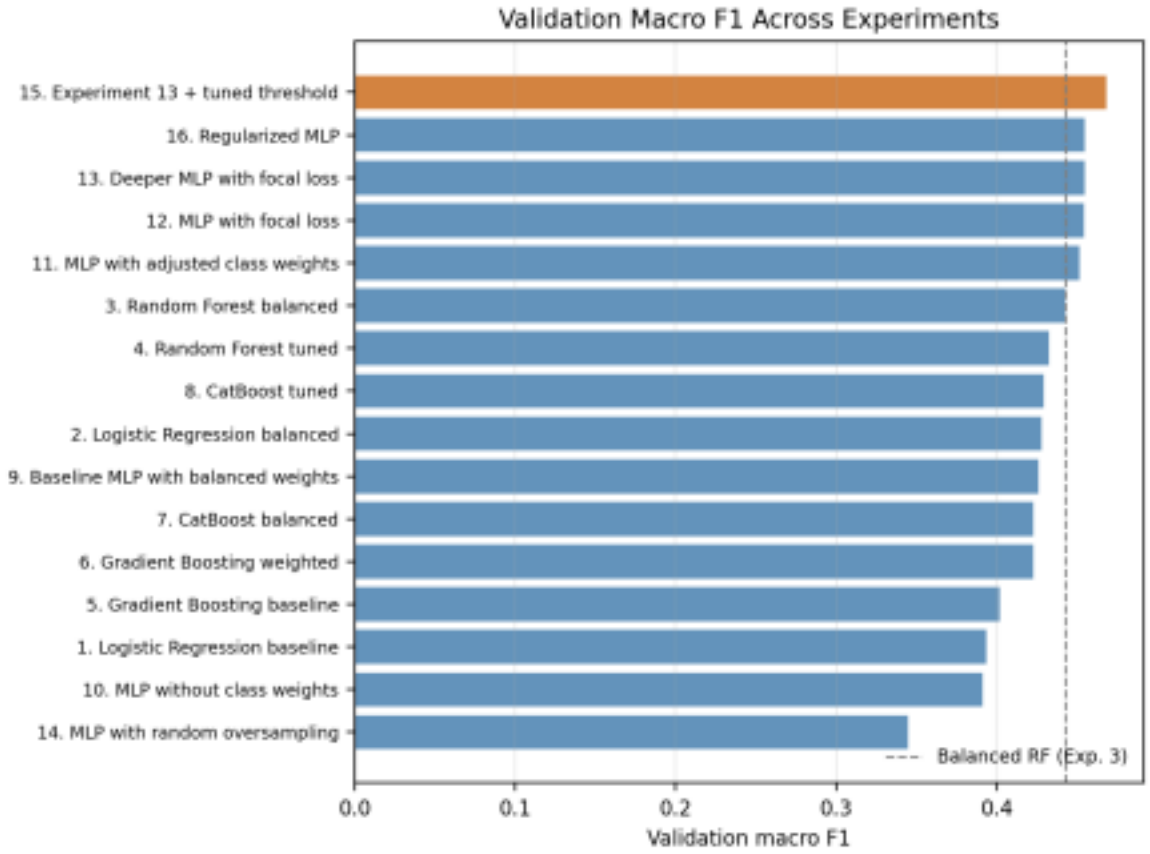


Fig. 2. Validation macro F1 across the experiment sequence. Experiment 15 is threshold tuning of Experiment 13, not a separately trained model.

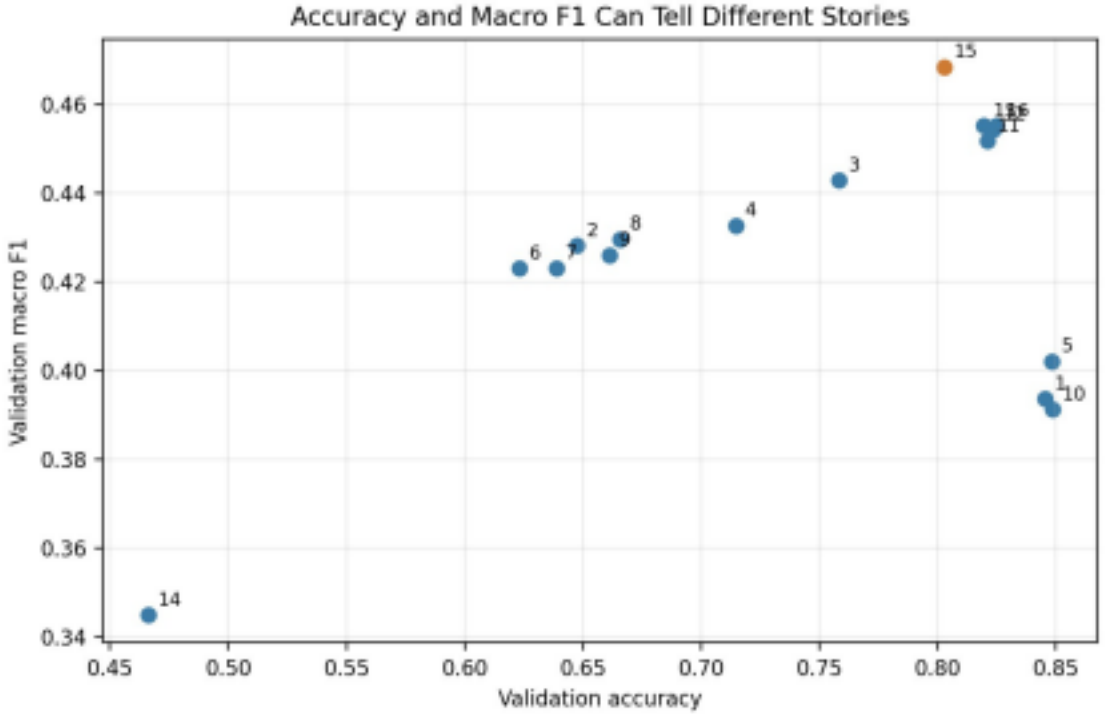


Fig. 3. Validation accuracy against macro F1. Labels are experiment numbers; Experiment 15 is a

decision-rule analysis.

D. Final evaluation on the untouched test set

The original held-out test partition contained 38,226 records: 84.55% no diabetes, 1.79% prediabetes, and 13.66% diabetes. The balanced Random Forest reached 76.12% accuracy and 0.4419 macro F1. The deeper MLP reached 82.15% accuracy and 0.4536 macro F1, close to its validation score of 0.4551, suggesting stable aggregate performance. However, its default predictions still contained no prediabetes cases.

Experiment 15 improved test macro F1 to 0.4614. Prediabetes precision was 3.90% and recall 6.28%, so most predicted prediabetes cases were false positives and most actual cases were still missed. Diabetes recall decreased from 52.95% to 48.54%. This is a small improvement in class balance, not reliable prediabetes detection.

TABLE III. FINAL TEST PERFORMANCE ON THE ORIGINAL UNSEEN TEST SPLIT

Test approach	Accuracy	Macro F1	Weighted F1	Prediabetes P	Prediabetes R	Diabetes R	Macro ROC AUC
Balance Random Forest (Exp. 3)	0.7612	0.4419	0.7848	0.0222	0.0117	0.6839	0.7583
Deep MLP(Exp. 13)	0.8215	0.4536	0.8218	0.0000	0.0000	0.5295	0.7781
MLP + tuned threshold (Exp. 15)	0.8046	0.4614	0.8146	0.0390	0.0628	0.4854	0.7781

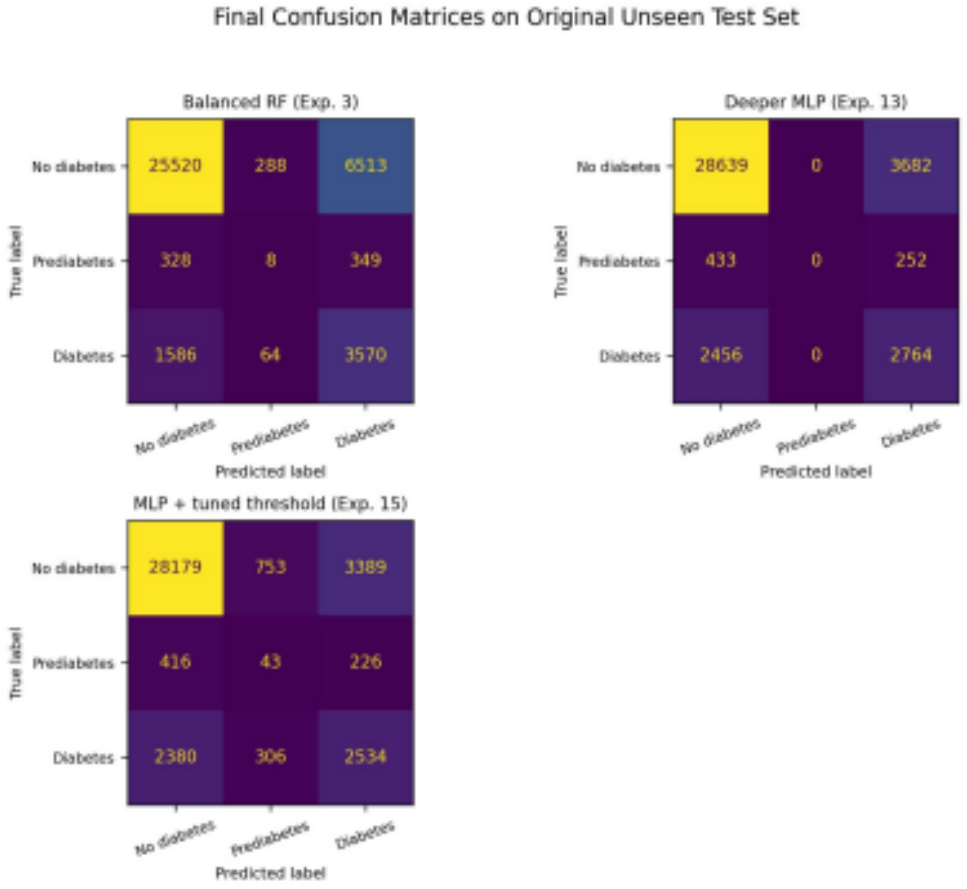


Fig. 4. Confusion matrices on the original unseen test set for the selected Random Forest, MLP, and validation-selected threshold rule.

E. Duplicate and conflicting-label sensitivity analysis

The sensitivity notebook tested whether duplicate removal and conflicting-label removal changed model results. It found 1,834 conflicting predictor patterns involving 6,120 records; after removing those patterns, 21,486 additional exact duplicate rows were removed. In total, 27,606 rows were removed, leaving 226,074. Class proportions shifted from 84.24% to 83.27% for no diabetes, from 1.83% to 1.90% for prediabetes, and from 13.93% to 14.83% for diabetes.

For a fair sensitivity comparison, all models were evaluated on the same cleaned test split. Random Forest macro F1 rose from 0.4377 for the original-trained model to 0.4419 after retraining on cleaned data. MLP macro F1 barely changed, from 0.4529 to 0.4530. Random Forest prediabetes recall stayed at 0.79%, and neither MLP predicted prediabetes. Cleaning did not resolve the main problem.

TABLE IV. SENSITIVITY COMPARISON: ORIGINAL-TRAINED AND CLEANED-TRAINED MODELS ON THE SAME CLEANED TEST SET

Model	Training data	Test accuracy	Test macro F1	Prediabetes precision	Prediabetes recall	Diabetes recall
Random Forest	Original	0.7445	0.4377	0.0180	0.0079	0.7048
Random Forest	Cleaned	0.7560	0.4419	0.0225	0.0079	0.6827
Deeper MLP	Original	0.8081	0.4529	0.0000	0.0000	0.5495
Deeper MLP	Cleaned	0.8099	0.4530	0.0000	0.0000	0.5408

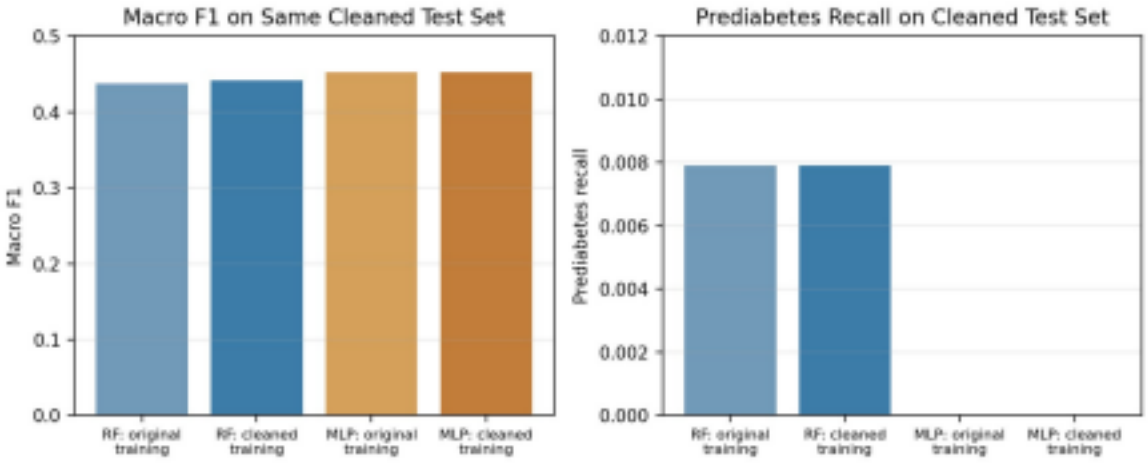


Fig. 5. Effect of cleaning training data on macro F1 and prediabetes recall when the test distribution is held constant.

V. DISCUSSION

A. Interpreting the model trade-offs

These results explain why macro F1 and class-specific recall were needed alongside accuracy. Because no-diabetes cases dominate, a model can reach high accuracy while missing prediabetes. The unweighted Logistic Regression, Gradient Boosting, and MLP baselines had around 84.5–84.9% accuracy but zero prediabetes recall. A high score alone would not support the project's health-awareness goal.

Class weighting improved minority-class recall in some experiments, including balanced Logistic Regression and weighted Gradient Boosting, but often reduced accuracy and increased false positives. No imbalance technique worked best for every class. The balanced Random Forest led the traditional models on validation macro F1, while the deeper focal-loss MLP led the trained neural networks. Both still detected prediabetes poorly.

Random oversampling repeated minority examples; it did not add new independent information. The resulting MLP performed poorly on the original validation distribution. This does not show that all resampling is ineffective, only that this configuration was unsuccessful. Future resampling must be restricted to training data and checked for plausible values for binary and ordinal features.

Experiment 15 chose a prediabetes threshold using validation data, then raised test macro F1 from 0.4536 to 0.4614. Yet 3.90% precision shows the false-positive burden. In a health-awareness tool, false positives may cause concern and missed cases may create false reassurance. The acceptable operating point would require clinical input, calibration, and prospective evaluation; the current threshold is not deployment-ready.

The test confusion matrices show the same problem in counts. The Random Forest correctly identified only 8 of 685 prediabetes cases, and the default MLP identified none. The threshold-tuned MLP identified 43 but also labelled many records from other classes as prediabetes. Prediabetes ROC-AUC around 0.63 to 0.68 indicates some ranking signal, but low Average Precision and hard-label scores show that this signal did not translate into reliable predictions. This is why precision-recall analysis matters for imbalanced data [14].

B. Learning curves and the bias-variance perspective

Training curves provide useful context but should not be overinterpreted. Experiment 10's training and validation losses decreased and stayed close, yet it never predicted prediabetes. Experiments 12 and 13 showed training loss decreasing after validation loss plateaued, suggesting diminishing gains and possible mild overfitting late in training. Experiment 16 had close curves after Batch Normalization and L2 regularization, but prediabetes recall remained zero. Regularization can limit complexity but cannot create missing predictive information.

Class-specific prediction bias is not the same as proving high estimator bias in a formal bias-variance decomposition. Likewise, a small training-validation loss gap does not prove the absence of variance. The defensible conclusion is that the selected MLP had similar validation and test macro F1, but still made systematic errors for the rare class. Absolute losses should not be compared directly across ordinary cross-entropy, weighted loss, and focal loss because the objectives differ.

C. What the cleaning sensitivity study means

The cleaning study asked whether exact duplicates and patterns with conflicting labels affected learning. Removing them changed class prevalence, with the share of diabetes increasing and prediabetes changing only slightly. Identical predictors paired with different labels may reflect missing features rather than incorrect records.

On the same cleaned test set, Random Forest macro F1 increased by 0.0042 after cleaning, while MLP macro F1 changed by only 0.0001. These small differences do not justify saying cleaning improved performance substantially. Removing conflicting-label rows may discard genuine respondents and affect

representativeness, so this should be treated as a sensitivity choice, not a correction that establishes which label was right.

D. Dataset and deployment limitations

The target is U.S. survey-reported status from 2015, not laboratory-confirmed diabetes or future disease onset. Self-report may include errors, and the dataset may not represent current populations or countries such as my country Rwanda. The class imbalance and conflicting labels also limit prediabetes detection. Therefore, the model should not be presented as a global predictor or diagnostic model.

A future model together with an application would require newer and representative data, independent validation in its intended population, probability calibration, privacy protections, communication of uncertainty, and clinical oversight. This study is an exploratory modelling step, not a deployable screening product.

E. Future iterations

Future work should target prediabetes discrimination rather than add complexity for its own sake. Group-aware cross-validation could test whether improvements are stable across splits; calibration could help interpret probability scores; and feature ablation could show which indicators provide useful signal. SMOTE or SMOTENC could be tested inside training folds only, with generated binary and ordinal values checked for plausibility. Deduplication should remain a separate sensitivity study, and external validation on current, representative data is necessary before considering practical use.

VI. CONCLUSION

This project compared traditional machine learning and deep learning for three-class diabetes-status classification using 21 indicators from a 2015 BRFSS-derived dataset. Eight traditional experiments and seven neural-network training experiments tested class weights, tree parameters, focal loss, architecture, regularization, and oversampling. Because prediabetes represented only 1.83% of the original data, macro F1 and class-level metrics were essential.

The balanced Random Forest was the strongest traditional candidate by validation macro F1 (0.4428). The deeper focal-loss MLP reached 0.4551 on validation and 0.4536 on the unseen test set. Threshold tuning raised test macro F1 to 0.4614, but prediabetes precision was 3.90% and recall 6.28%. Cleaning caused only small changes and did not improve MLP prediabetes recall.

My motivation was to explore whether a future mobile tool empowered by Machine learning model could encourage people to seek professional screening. This study is only an initial investigation, not a diagnostic application. Better prediabetes detection, representative data, external validation, calibration, and clinical collaboration are needed before the idea can be evaluated for real-world use.

REFERENCES

[1] World Health Organization, "Diabetes," Fact Sheet, Nov. 14, 2024. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/diabetes.Accessed: Oct. 18, 2026.

[2] Centers for Disease Control and Prevention, "Prediabetes: Could It Be You?," updated Feb. 17, 2026. [Online]. Available: https://www.cdc.gov/diabetes/communication-resources/prediabetes-statistics.html. Accessed: Oct. 18, 2026.

[3] Centers for Disease Control and Prevention, "2015 BRFSS Survey Data and Documentation." [Online]. Available: https://www.cdc.gov/brfss/annual_data/annual_2015.html. Accessed: Oct. 18, 2026.

[4] A. Teboul, "Diabetes Health Indicators Dataset," Kaggle. [Online]. Available: https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset. Accessed: Oct. 18, 2026.

[5] J. Hippisley-Cox, C. Coupland, J. Robson, A. Sheikh, and P. Brindle, "Predicting risk of type 2 diabetes in England and Wales: Prospective derivation and validation of QDScore," BMJ, vol. 338, Art. no. b880, 2009, doi: 10.1136/bmj.b880.

[6] D. Noble, R. Mathur, T. Dent, C. Meads, and T. Greenhalgh, "Risk models and scores for type 2 diabetes: Systematic review," BMJ, vol. 343, Art. no. d7163, 2011, doi: 10.1136/bmj.d7163.

[7] G. S. Collins, S. Mallett, O. Omar, and L.-M. Yu, "Developing risk prediction models for type 2 diabetes: A systematic review of methodology and reporting," BMC Medicine, vol. 9, Art. no. 103, 2011, doi: 10.1186/1741-7015-9-103.

[8] V. Chang, M. A. Ganatra, K. Hall, L. Golightly, and Q. A. Xu, "An assessment of machine learning models and algorithms for early prediction and diagnosis of diabetes using health indicators," Healthcare Analytics, vol. 2, Art. no. 100118, 2022, doi: 10.1016/j.health.2022.100118.

[9] K. De Silva, W. K. Lee, A. Forbes, R. T. Demmer, C. Barton, and J. Enticott, "Use and performance of machine learning models for type 2 diabetes prediction in community settings: A systematic review and meta-analysis," International Journal of Medical Informatics, vol. 143, Art. no. 104268, 2020, doi: 10.1016/j.ijmedinf.2020.104268.

[10] G. S. Collins, J. B. Reitsma, D. G. Altman, and K. G. M. Moons, "Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): The TRIPOD statement," BMJ, vol. 350, Art. no. g7594, 2015, doi: 10.1136/bmj.g7594.

[11] H. He and E. A. Garcia, "Learning from imbalanced data," IEEE Transactions on Knowledge and Data Engineering, vol. 21, no. 9, pp. 1263–1284, Sep. 2009, doi: 10.1109/TKDE.2008.239.

[12] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," Journal of Artificial Intelligence Research, vol. 16, pp. 321–357, 2002, doi: 10.1613/jair.953.

[13] T. Fawcett, "An introduction to ROC analysis," Pattern Recognition Letters, vol. 27, no. 8, pp. 861–874, 2006, doi: 10.1016/j.patrec.2005.10.010.

[14] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," PLoS ONE, vol. 10, no. 3, Art. no. e0118432, 2015, doi: 10.1371/journal.pone.0118432.

[15] L. Breiman, "Random forests," Machine Learning, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[16] J. H. Friedman, “Greedy function approximation: A gradient boosting machine,” The Annals of Statistics, vol. 29, no. 5, pp. 1189–1232, 2001, doi: 10.1214/aos/1013203451.

[17] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, “CatBoost: Unbiased boosting with categorical features,” in Advances in Neural Information Processing Systems, vol. 31, 2018, pp. 6638–6648. [18] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.

APPENDIX A. ADDITIONAL EXPERIMENT FIGURES

The following figures provide supporting visual evidence for the experiment discussion. In the final submission, the notebook and W&B project links should point to the complete reproducible records. Captions identify the purpose of each figure so that the graphics can be referenced from the report text.

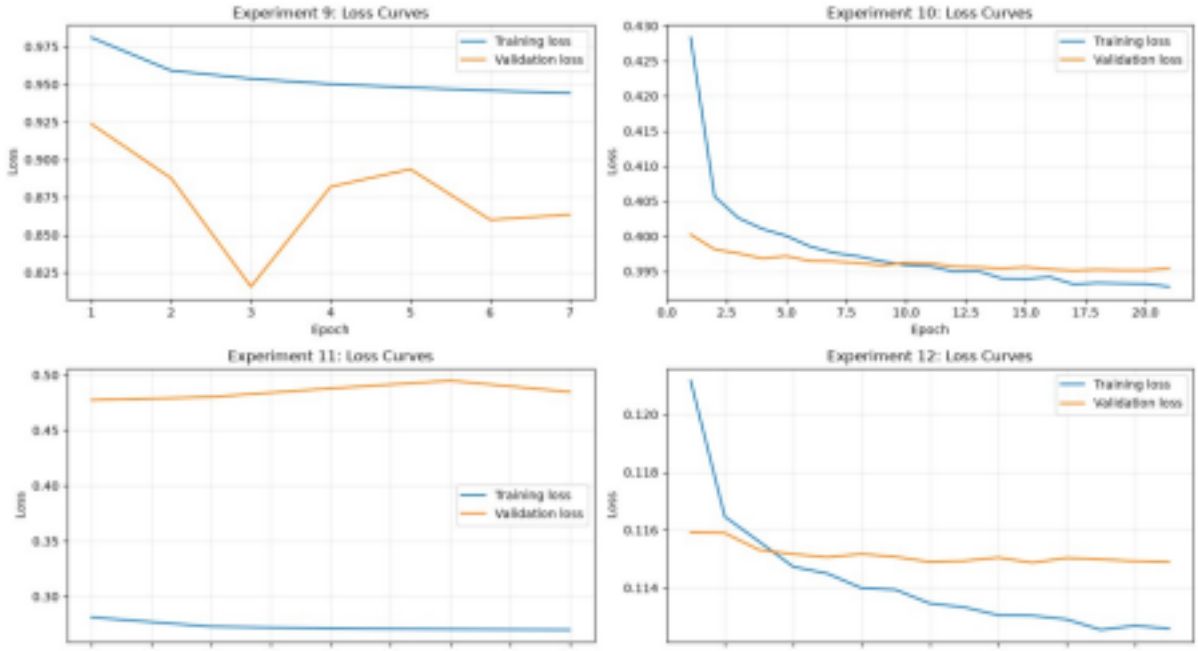


Fig. A1. Training and validation loss curves for Experiments 9–12, as recorded in the notebook.

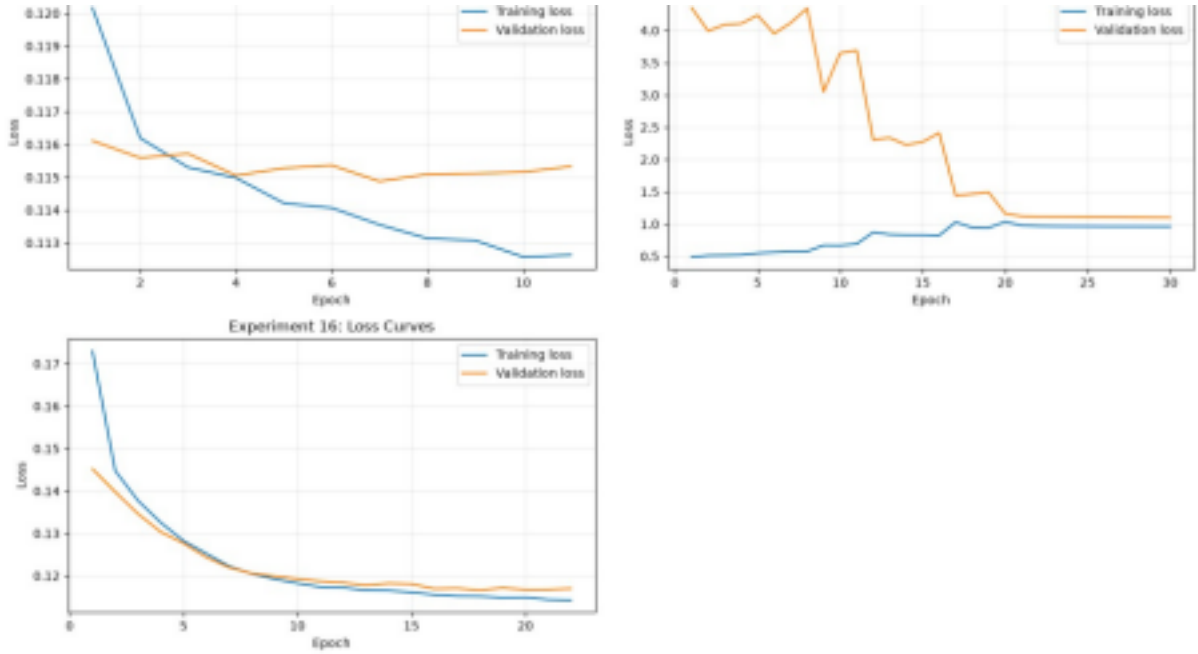


Fig. A2. Training and validation loss curves for Experiments 13, 14, and 16. Experiment 14 displays a substantially different and unstable validation-loss scale.

APPENDIX B. EXPERIMENT CONFIGURATION SUMMARY

Table B1 summarizes the major experiment variations. Exact run configuration and metric records are available in the notebooks and the linked WandB project. The validation split was used for model comparison; the original test split was used once for the final comparison of the selected models; the

cleaned-test comparison belongs to the separate sensitivity notebook.

TABLE B1. SUMMARY OF EXPERIMENT CONFIGURATIONS

Experiment	Model	Main configuration variation
1	Logistic Regression	max_iter=1000
2	Logistic Regression	class_weight=balanced; max_iter=1000
3	Random Forest	200 trees; max_depth=None; min_samples_leaf=2; balanced
4	Random Forest	200 trees; max_depth=20; min_samples_leaf=5; balanced
5	Gradient Boosting	100 estimators; learning_rate=0.1; depth=3
6	Gradient Boosting	baseline settings + balanced sample weights
7	CatBoost	500 iterations; depth=6; learning_rate=0.05; auto_class_weights=Balanced
8	CatBoost	500 iterations; depth=8; learning_rate=0.05; auto_class_weights=Balanced
9	MLP	[128,64]; dropout [0.3,0.2]; balanced class weights; Adam 0.001
10	MLP	[128,64]; dropout [0.3,0.2]; no class weights; Adam 0.001
11	MLP	[128,64]; inverse-square-root normalized class weights
12	MLP focal loss	[128,64]; gamma=2; inverse-square-root weights
13	Deeper MLP focal loss	[256,128,64]; dropout [0.3,0.3,0.2]; gamma=2
14	MLP oversampling	[128,64]; random oversampling on training data only

15	Threshold analysis	Experiment 13 probabilities; threshold=0.24203; no retraining
16	Regularized MLP	[256,128,64]; Batch Normalization; L2=1e-4; focal loss